written $f(1) + f(2) + \cdots + f(n-1) + f(n)$. Within a sentence, Sigma notation is typeset as $\sum_{i=1}^{n} f(i)$.

Given a summation, you often wish to replace it with an algebraic equation with the same value as the summation. This is known as a **closed-form solution**, and the process of replacing the summation with its closed-form solution is known as **solving** the summation. For example, the summation $\sum_{i=1}^{n} 1$ is simply the expression "1" summed n times (remember that i ranges from 1 to n). Because the sum of n 1s is n, the closed-form solution is n. The following is a list of useful summations, along with their closed-form solutions.

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}. \tag{2.1}$$

$$\sum_{i=1}^{n} i^2 = \frac{2n^3 + 3n^2 + n}{6} = \frac{n(2n+1)(n+1)}{6}. \tag{2.2}$$

$$\sum_{i=1}^{\log n} n = n \log n. \tag{2.3}$$

$$\sum_{i=0}^{\infty} a^i = \frac{1}{1-a} \text{ for } 0 < a < 1. \tag{2.4}$$

$$\sum_{i=0}^{n} a^i = \frac{a^{n+1} - 1}{a - 1} \text{ for } a \neq 1. \tag{2.5}$$

As special cases to Equation 2.5,

$$\sum_{i=1}^{n} \frac{1}{2^i} = 1 - \frac{1}{2^n}, \tag{2.6}$$

and

$$\sum_{i=0}^{n} 2^i = 2^{n+1} - 1. \tag{2.7}$$

As a corollary to Equation 2.7,

$$\sum_{i=0}^{\log n} 2^i = 2^{\log n + 1} - 1 = 2n - 1. \tag{2.8}$$

Finally,

$$\sum_{i=1}^{n} \frac{i}{2^i} = 2 - \frac{n+2}{2^n}. \tag{2.9}$$

The sum of reciprocals from 1 to n, called the **Harmonic Series** and written $\mathcal{H}_n$, has a value between $\log_e n$ and $\log_e n + 1$. To be more precise, as n grows, the